SIMATS ENGINEERING

CO3- AT2 - Design Task - Medium

COURSE NAME: Advance Wireless Communication Systems /

5G / 6G Communication Systems

COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 3 - *Design spectrum access strategies, waveform generation methods, and multiple access techniques for efficient wireless transmission systems.* (BL6)

Assessment Tool Weightage: 35%

Short description about assessment: Students design a system / model based on given requirements to assess creativity and application skills.

Dr.

QUESTIONS: 1 (1 X 100 = 100)   Total Marks: 100 Duration: 3hrs

1. Design a system architecture for Multi-access Edge Computing to reduce latency in smart traffic applications. (BL6)
2. Develop a cloud-edge communication model for wireless healthcare monitoring. (BL6)
3. Design a functional architecture for real-time edge data processing in industrial automation. (BL6)
4. Propose a cloud-based wireless framework for smart surveillance systems. (BL6)
5. Design an edge AI workflow for intelligent wireless data filtering. (BL6)
6. Develop a task offloading strategy between edge and cloud in 5G applications. (BL6)
7. Design a low-latency communication architecture for edge-assisted smart city services. (BL6)
8. Propose a scalable edge-cloud framework for wireless sensor networks. (BL6)
9. Design a Cloud Radio Access Network model for dense urban deployment. (BL6)
10. Develop a functional split model between centralized and distributed radio units. (BL6)
11. Design a virtualized wireless access framework for next-generation communication. (BL6)
12. Propose a data flow model for cloud-based radio resource allocation. (BL6)
13. Design a centralized traffic control model using Cloud RAN principles. (BL6)
14. Develop a network management architecture for cloud-assisted wireless systems. (BL6)
15. Design a wireless service framework integrating virtual network functions. (BL6)
16. Propose an optimized fronthaul communication model. (BL6)
17. Design a satellite-assisted communication model using Low Earth Orbit satellite for remote wireless coverage. (BL6)

18. Develop a comparative architecture for LEO, MEO, and GEO backhaul systems. (BL6)
19. Design a spectrum sharing model between LTE and 5G users. (BL6)
20. Propose a frequency allocation strategy for mixed wireless services. (BL6)
21. Design a dynamic spectrum access framework for urban wireless systems. (BL6)
22. Develop a communication architecture combining terrestrial and satellite networks. (BL6)
23. Design a spectrum efficiency improvement model for dense communication zones. (BL6)
24. Propose a channel reuse plan for multi-user wireless communication. (BL6)
25. Design an NB-IoT architecture for smart utility monitoring. (BL6)
26. Develop an LTE-M communication workflow for wearable monitoring systems. (BL6)
27. Design a low-power communication model for battery-operated IoT devices. (BL6)
28. Propose a device communication plan for long-life sensor networks. (BL6)
29. Design a 5G-enabled IoT framework for campus automation. (BL6)
30. Develop an energy-efficient wireless communication strategy for sensor nodes. (BL6)
31. Design an IoT communication flow for periodic environmental sensing. (BL6)
32. Propose a scalable wireless architecture for smart connected devices. (BL6)
33. Design a smart city communication framework for traffic monitoring and surveillance. (BL6)
34. Develop an energy monitoring communication model for public infrastructure. (BL6)
35. Design a smart factory information flow for predictive maintenance. (BL6)
36. Propose a robotic automation communication framework for industrial control. (BL6)
37. Design a remote patient monitoring communication architecture. (BL6)
38. Develop a wireless communication workflow for AI-assisted healthcare diagnosis. (BL6)
39. Design a smart irrigation communication model for crop monitoring. (BL6)
40. Propose a livestock tracking communication architecture using wireless sensors. (BL6)

Code of Conduct:

I ______________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Design Task					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Design	20%	Demonstrates deep understanding of design objectives, constraints, and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Lacks understanding of the design context
Evaluation of Design	25%	Provides in-depth critique identifying strengths, weaknesses, and design gaps	Identifies key issues with reasonable analysis	Limited critique; mostly descriptive feedback	No meaningful critique provided
Justification	20%	Supports critique with strong reasoning, evidence, and relevant principles	Provides justification with some supporting evidence	Weak or unclear justification	No justification for feedback
Suggestions for Improvement	20%	Offers innovative, practical, and well-justified improvement suggestions	Provides useful suggestions with minor gaps	Suggestions are limited or partially relevant	No constructive suggestions provided
Communication	15%	Communicates feedback clearly, professionally, and engages actively in discussion	Communicates well with minor issues	Communication lacks clarity or engagement	Poor communication and minimal participation

Task:

- To design a reliable communication architecture for real-time remote monitoring of patient health parameters.
- To acquire physiological parameters such as ECG, heart rate, SpO_2, blood pressure, and body temperature using appropriate sensors.
- To transmit patient data from the patient-side device to a remote healthcare monitoring station.
- To evaluate the effect of different SNR levels and communication conditions on reliable patient-data transmission.
- To provide timely alerts to healthcare professionals when monitored parameters exceed predefined threshold values.

Description

- The system enables continuous remote monitoring of a patient's physiological parameters using connected medical sensors.
- Sensors collect important health parameters such as ECG, heart rate, SpO_2, body temperature, and blood pressure.
- The collected physiological signals are sent to an ESP32 or microcontroller-based edge device for initial processing.
- The edge device performs signal conditioning, filtering, digitization, and data processing before transmission.
- Processed patient data is transmitted through a wireless communication network such as Wi-Fi, Bluetooth, LoRa, 4G, or 5G.
- A gateway or cloud server receives the transmitted data and provides storage, processing, and visualization.
- Healthcare professionals can access the patient's information through a remote monitoring dashboard.
- The system can identify abnormal health conditions using predefined threshold values and generate alerts.
- MATLAB can be used to evaluate communication performance based on BER, throughput, latency, SNR, and packet delivery ratio.

Key Parameters

Parameter	Value	Purpose
Number of patients	10	Number of monitored users
Sampling frequency	250 Hz	Physiological signal sampling
Data resolution	16 bits	ADC resolution
Channel bandwidth	1 MHz	Wireless channel bandwidth
Carrier frequency	2.4 GHz	Example wireless carrier
Modulation	QPSK	Digital modulation
Data rate	1 Mbps	Patient-data transmission
SNR	0–20 dB	Channel quality
Packet size	256 bytes	Data packet size
Packet interval	1 s	Transmission interval
Packet-loss threshold	5%	Communication reliability
Latency target	<100 ms	Near-real-time monitoring
ECG sampling rate	250 Hz	ECG acquisition
ECG resolution	16 bits	ECG quantization
Heart-rate range	60–100 BPM	Example normal range
SpO_2 threshold	>95%	Example normal range
Temperature	36.5–37.5 °C	Example normal range

Matlab code

```
clc;
clear;
close all;

%% REMOTE PATIENT MONITORING COMMUNICATION SYSTEM

%% Parameters
numPatients = 10;          % Number of patients
fs = 250;                  % Sampling frequency (Hz)
bitsPerSample = 16;        % ADC resolution
packetSize = 256;          % Packet size in bytes
SNRdB = 0:2:20;            % SNR range
dataRate = 1e6;            % Communication data rate
latencyProcessing = 20e-3; % Processing latency
propagationDelay = 5e-3;   % Propagation delay

%% Generate physiological data

t = 0:1/fs:1-1/fs;

% Example ECG-like signal
ecg = 1.2*sin(2*pi*1.2*t) + ...
     0.25*sin(2*pi*2.4*t) + ...
     0.1*randn(size(t));

% Heart rate
heartRate = 60 + 5*sin(2*pi*0.1*t);

% SpO2
SpO2 = 97 + 0.5*sin(2*pi*0.05*t);

% Temperature
temperature = 36.8 + 0.2*sin(2*pi*0.02*t);

%% Display physiological signals

figure;
```

```
subplot(3,1,1);
plot(t,ecg);
xlabel('Time (s)');
ylabel('Amplitude');
title('ECG Signal');
grid on;

subplot(3,1,2);
plot(t,heartRate);
xlabel('Time (s)');
ylabel('BPM');
title('Heart Rate');
grid on;

subplot(3,1,3);
plot(t,SpO2);
xlabel('Time (s)');
ylabel('SpO2 (%)');
title('Blood Oxygen Level');
grid on;

%% Convert patient data to binary

numBits = 10000;

txBits = randi([0 1],numBits,1);

%% QPSK Modulation

% Make number of bits even
if mod(length(txBits),2) ~= 0
    txBits = [txBits;0];
end

bitPairs = reshape(txBits,2,[]).';
```

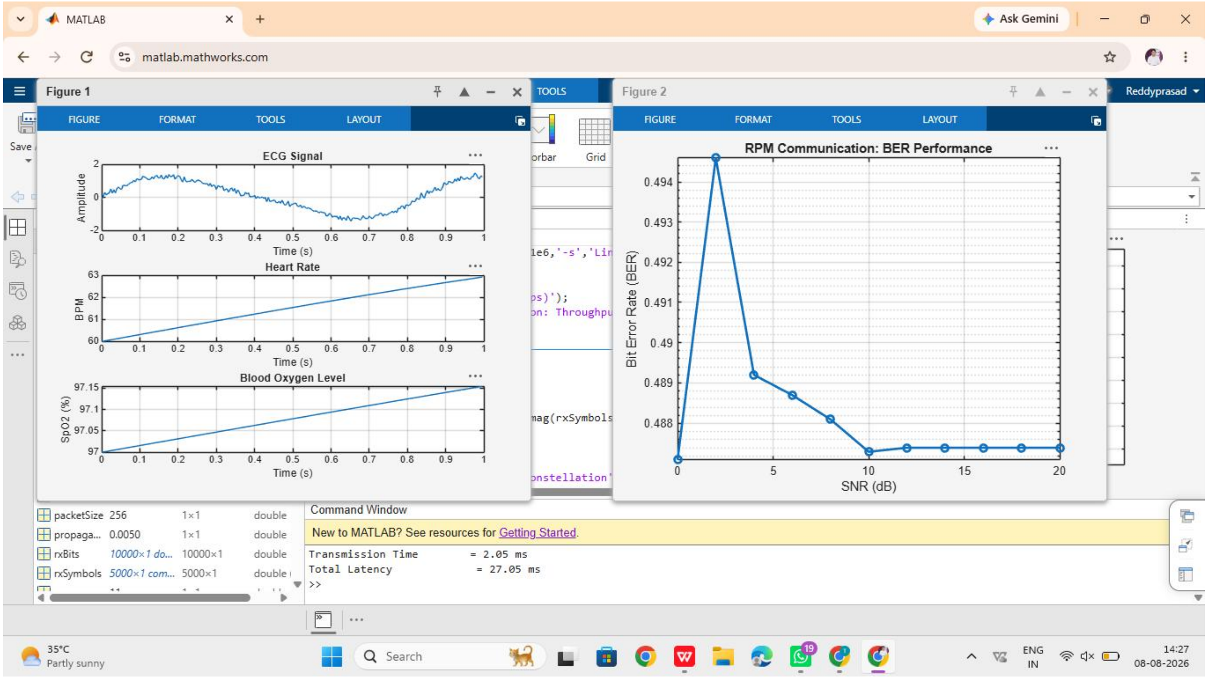
```
% QPSK mapping
symbols = zeros(size(bitPairs,1),1);

for k = 1:size(bitPairs,1)

    if bitPairs(k,1)==0 && bitPairs(k,2)==0
        symbols(k) = 1 + 1i;
    elseif bitPairs(k,1)==0 && bitPairs(k,2)==1
        symbols(k) = -1 + 1i;
    elseif bitPairs(k,1)==1 && bitPairs(k,2)==1
        symbols(k) = -1 - 1i;
    else
        symbols(k) = 1 - 1i;
    end

end

% Normalize power
symbols = symbols/sqrt(2);

%% BER calculation

BER = zeros(size(SNRdB));
throughput = zeros(size(SNRdB));

for s = 1:length(SNRdB)

    %% AWGN Channel

    SNRlinear = 10^(SNRdB(s)/10);

    noisePower = 1/SNRlinear;

    noise = sqrt(noisePower/2) * ...
        (randn(size(symbols)) + ...
         1i*randn(size(symbols)));
```

```
    rxSymbols = symbols + noise;

    %% QPSK Demodulation

    rxBits = zeros(length(txBits),1);

    for k = 1:length(rxSymbols)

        if real(rxSymbols(k)) >= 0
            rxBits(2*k-1) = 0;
        else
            rxBits(2*k-1) = 1;
        end

        if imag(rxSymbols(k)) >= 0
            rxBits(2*k) = 0;
        else
            rxBits(2*k) = 1;
        end

    end

    %% BER

    BER(s) = sum(txBits ~= rxBits)/length(txBits);

    %% Throughput

    throughput(s) = dataRate*(1-BER(s));

end

%% Latency

transmissionTime = (packetSize*8)/dataRate;

totalLatency = transmissionTime + ...
```

```
        latencyProcessing + ...
        propagationDelay;

%% Display results

fprintf('\nREMOTE PATIENT MONITORING RESULTS\n');
fprintf('-----------------------------------\n');

fprintf('Number of Patients       = %d\n',numPatients);
fprintf('Sampling Frequency       = %d Hz\n',fs);
fprintf('Packet Size              = %d bytes\n',packetSize);
fprintf('Data Rate                = %.2f Mbps\n',dataRate/1e6);
fprintf('Processing Latency       = %.2f ms\n',latencyProcessing*1000);
fprintf('Propagation Delay        = %.2f ms\n',propagationDelay*1000);
fprintf('Transmission Time        = %.2f ms\n',transmissionTime*1000);
fprintf('Total Latency            = %.2f ms\n',totalLatency*1000);

%% BER plot

figure;

semilogy(SNRdB,BER,'-o','LineWidth',2);

xlabel('SNR (dB)');
ylabel('Bit Error Rate (BER)');
title('RPM Communication: BER Performance');
grid on;

%% Throughput plot

figure;

plot(SNRdB,throughput/1e6,'-s','LineWidth',2);

xlabel('SNR (dB)');
ylabel('Throughput (Mbps)');
title('RPM Communication: Throughput vs SNR');
```

```
grid on;

%% QPSK Constellation

figure;

plot(real(rxSymbols),imag(rxSymbols),'.');

xlabel('In-phase');
ylabel('Quadrature');
title('Received QPSK Constellation');
grid on;
```

Output



Formula

Data generation rate:

$$\boxed{R_{data} = N_s \times N_{bits}}$$

where N_s is the number of samples per second.

Transmission latency:

$$T_{tx}=RL$$

where L is packet length in bits and R is data rate.

Total latency:

$$T_{total}=T_{tx}+T_{processing}+T_{propagation}$$

Throughput:

$$Throughput=R_{data}(1-BER)$$

Packet delivery ratio:

$$PDR=N_{transmitted}N_{received}\times 100$$

Summary

- The proposed Remote Patient Monitoring (RPM) Communication Architecture enables continuous monitoring of patient health parameters through wireless communication.
- Physiological data such as ECG, heart rate, SpO_2, blood pressure, and temperature are collected using sensors.
- QPSK modulation and AWGN channel modeling can be used in MATLAB to evaluate the communication link.
- The system performance is analyzed using BER, throughput, latency, and packet delivery ratio.
- The architecture supports real-time monitoring and early alerts, helping healthcare professionals respond quickly to abnormal patient conditions.
- The proposed system can be extended to IoT, cloud-based healthcare, 5G, and wearable medical monitoring applications.